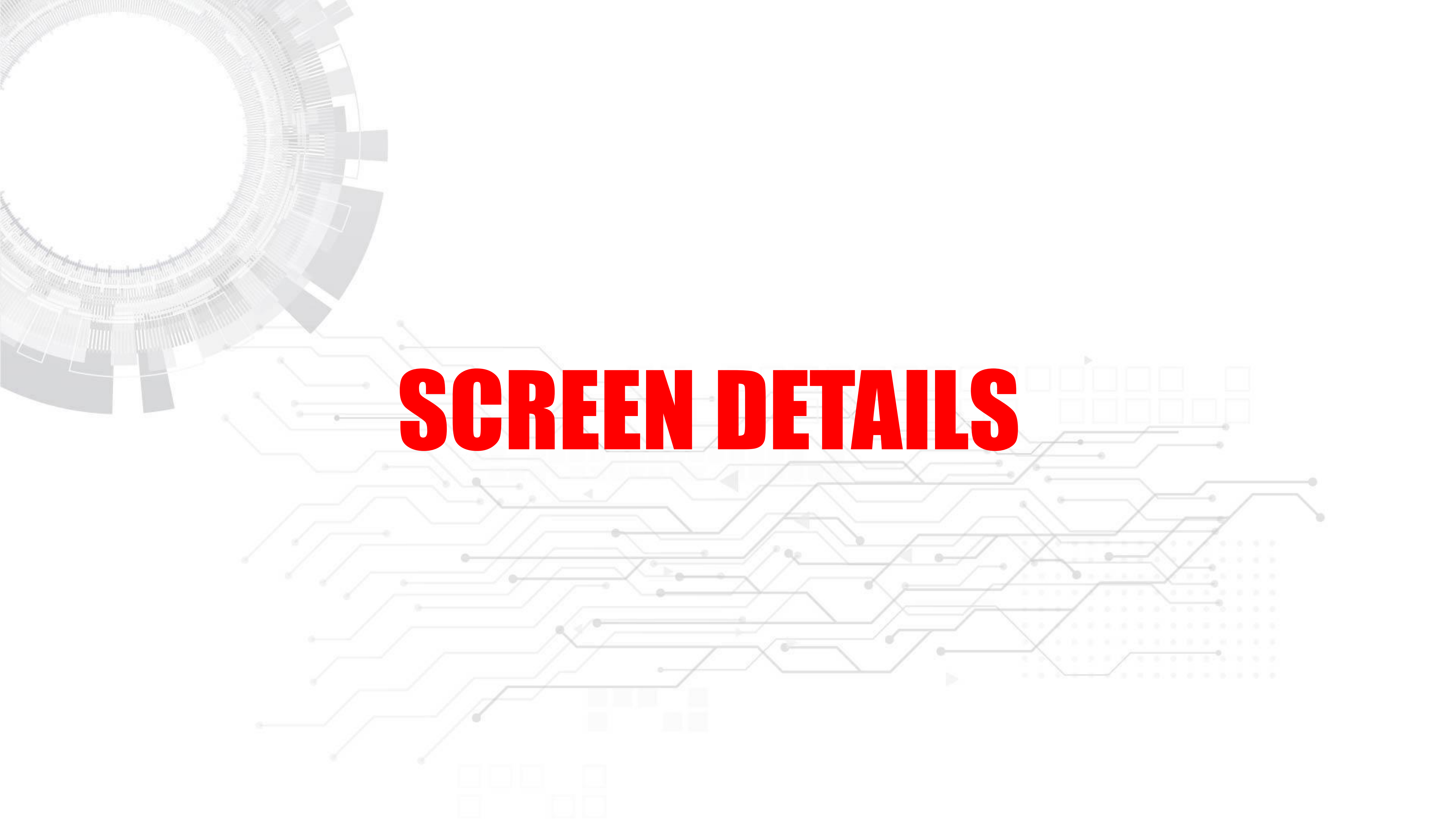


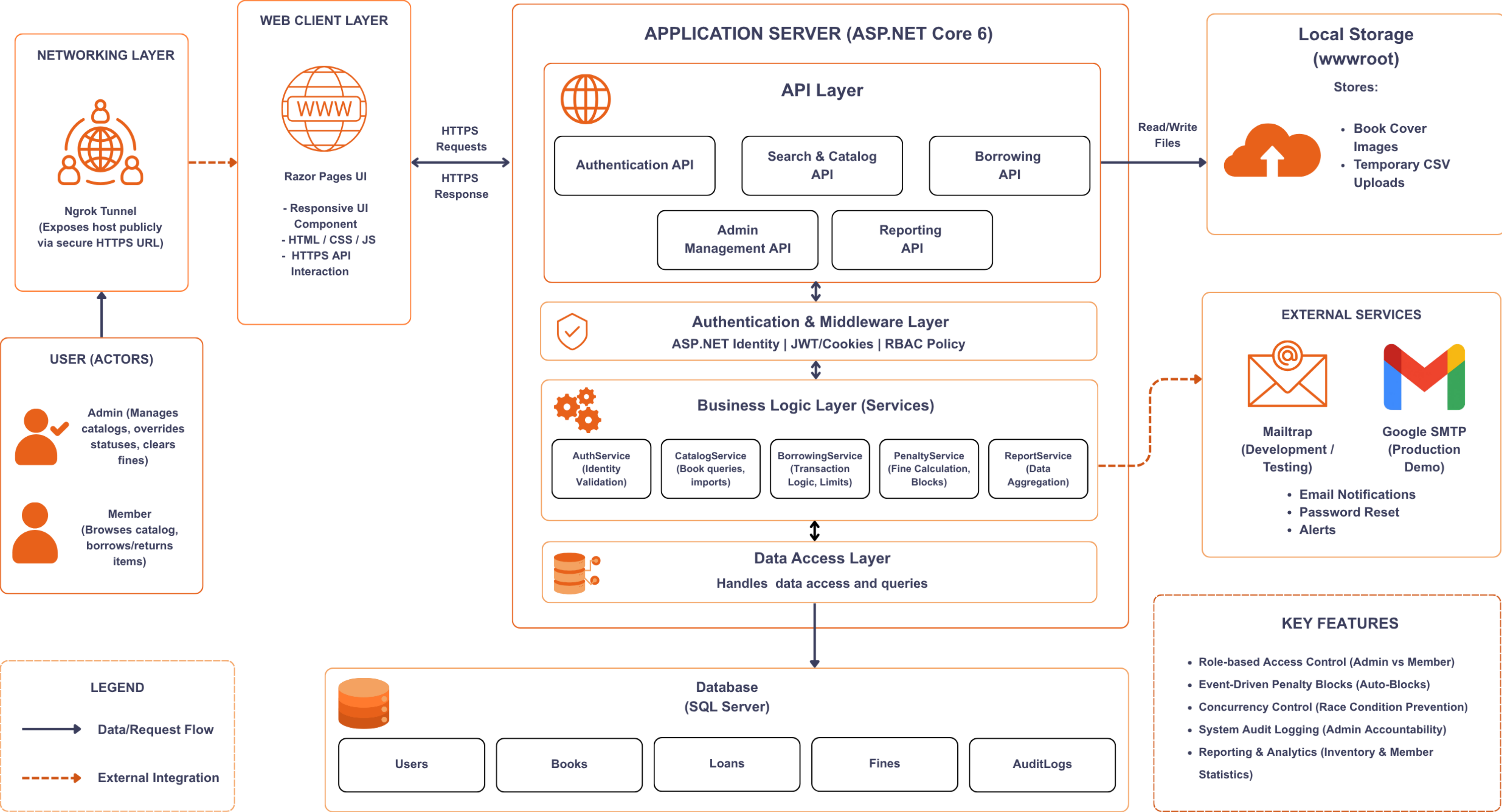
High-Level Design
Sharp Mindz - Project Minerva

Revision History

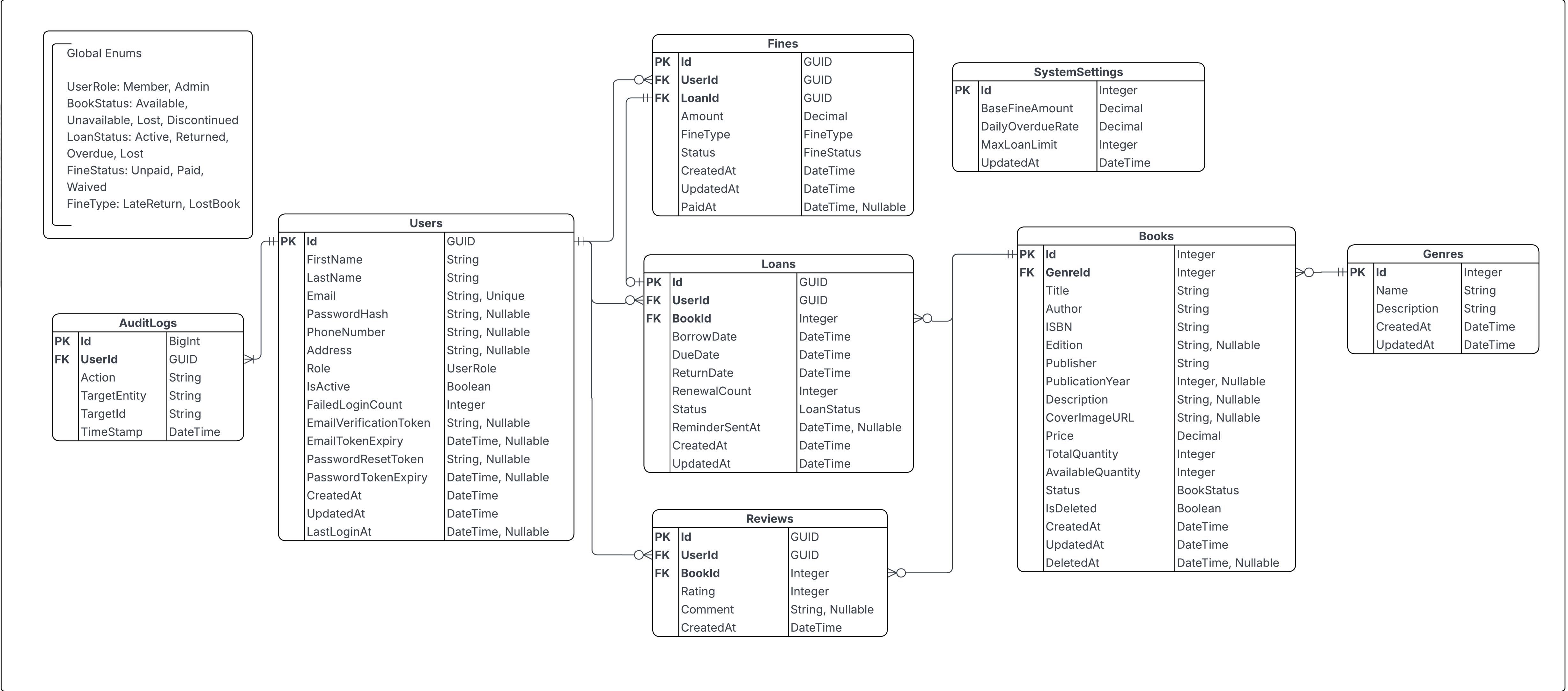


SCREEN DETAILS

Architectural Diagram



Entity Relationship Diagram (ERD)



Screen List

SCREEN	DESCRIPTION	RIGHTS / PERMISSIONS		REMARKS
		ADMIN	USER	
SCR-01 Landing Page	System introduction, feature overview, public search, and CTAs	x	x	
SCR-02 Login Page	Authenticates members and admins via email/password	x	x	
SCR-03 Signup Page	Allows new members to register with name, email, and password	x	x	Sends confirmation email on success
SCR-04 Forgot Password Page	Request password reset via email input	x	x	Triggers Password Reset Email
SCR-05 Change Password Page	Upon clicking the link from the email, open the change password page	x	x	
SCR-06 Member Catalog Dashboard	Home screen for logged-in members with search, genre chips, and book grid		x	

Screen List

SCREEN	DESCRIPTION	RIGHTS / PERMISSIONS		REMARKS
		ADMIN	USER	
SCR-07 Book Details Page	Displays full book metadata, description, and live availability		x	
SCR-08 Loans & History Page	Tabbed view for Current Loans and Borrowing History		x	
SCR-08 User Profile & Settings	View/edit profile info, change password, and view accrued fines		x	
SCR-10 Admin Dashboard	Command center overview with KPI widgets	x		
SCR-11 Book Management List	Searchable master inventory table with row-level actions	x		Supports bulk import and catalog export (CSV)
SCR-12 Add / Edit Book Form	Form for text entry and cover image upload for book records	x		

Screen List

SCREEN	DESCRIPTION	RIGHTS / PERMISSIONS		REMARKS
		ADMIN	USER	
SCR-13 System Audit Trail	Read-only log ledger of all administrative actions	x		
SCR-14 Reports	Generate Borrowing, Inventory, or Member Activity reports	x		
SCR-15 Member Management	Search for a suspended member, view their outstanding balance, and execute the fine clearance override.	x		

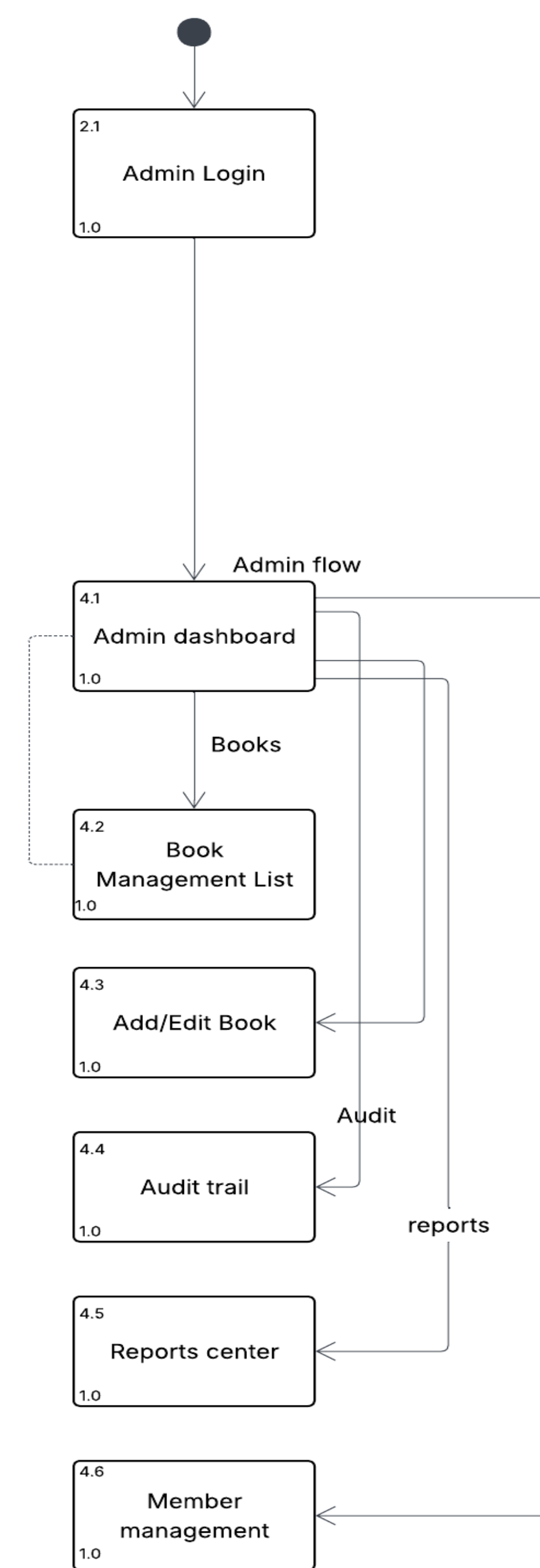
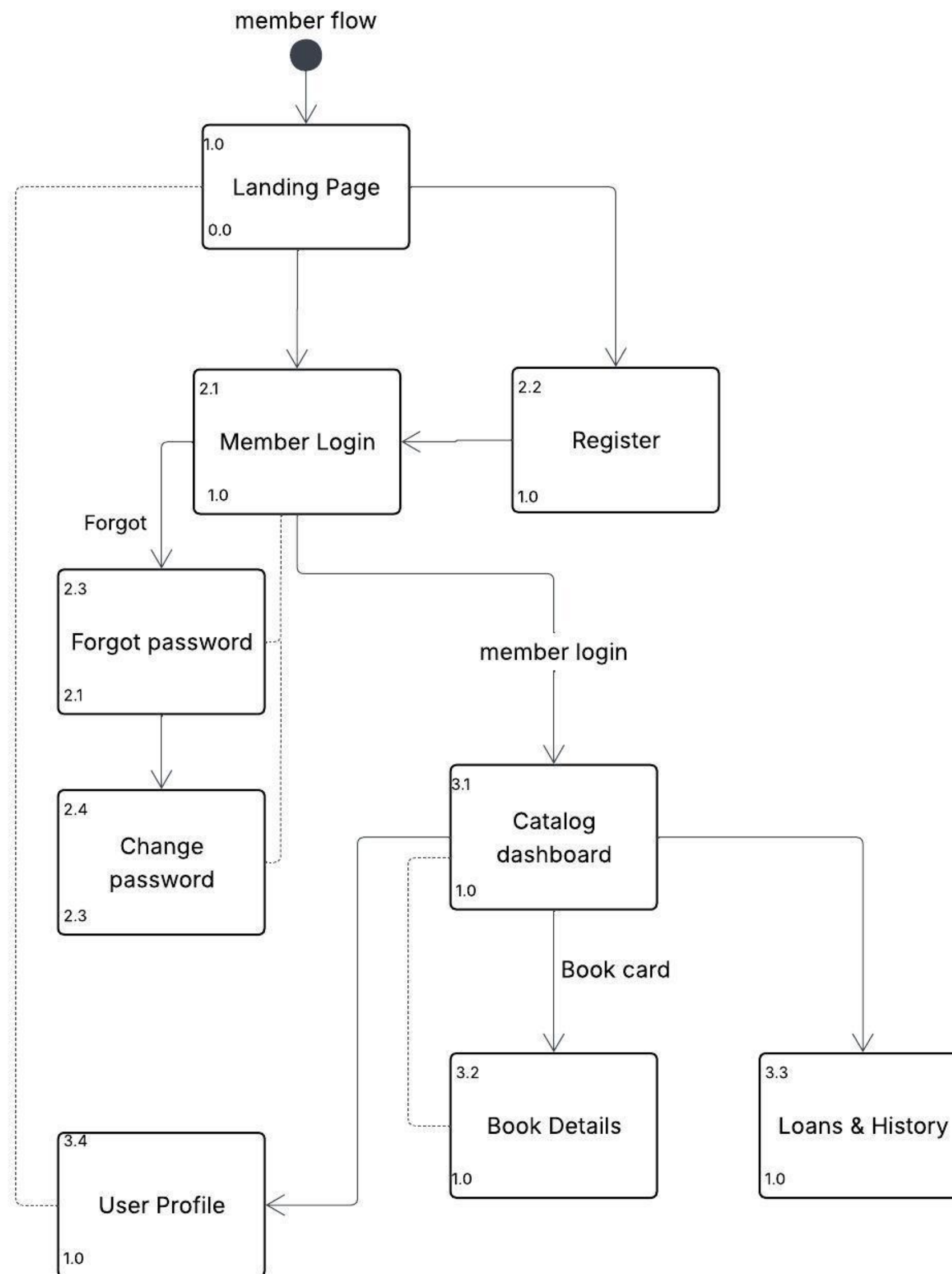
Modal List List

SCREEN	DESCRIPTION	RIGHTS / PERMISSIONS		REMARKS
		ADMIN	USER	
MDL-01 Borrow Confirmation	Prompts the user to confirm they want to borrow the selected book.		x	Triggered from SCR-07 Book Details
MDL-02 Return Confirmation	Prompts the user to confirm they are returning the book.		x	Triggered from SCR-08 Loans & History
MDL-03 Logout Confirmation	Confirms the user or admin wants to end their session.	x	x	
MDL-04 Delete Book Warning	A destructive action warning before permanently removing a book from the DB.	x		Triggered from SCR-11 Book Management
MDL-05 Bulk Import Upload	A small dialog containing a drag-and-drop zone or file picker for the CSV file.	x		Triggered from SCR-11 Book Management
MDL-06 Status Override	A dropdown dialog allowing admins to change a book to "Lost", "Damaged", or "Archived".	x		

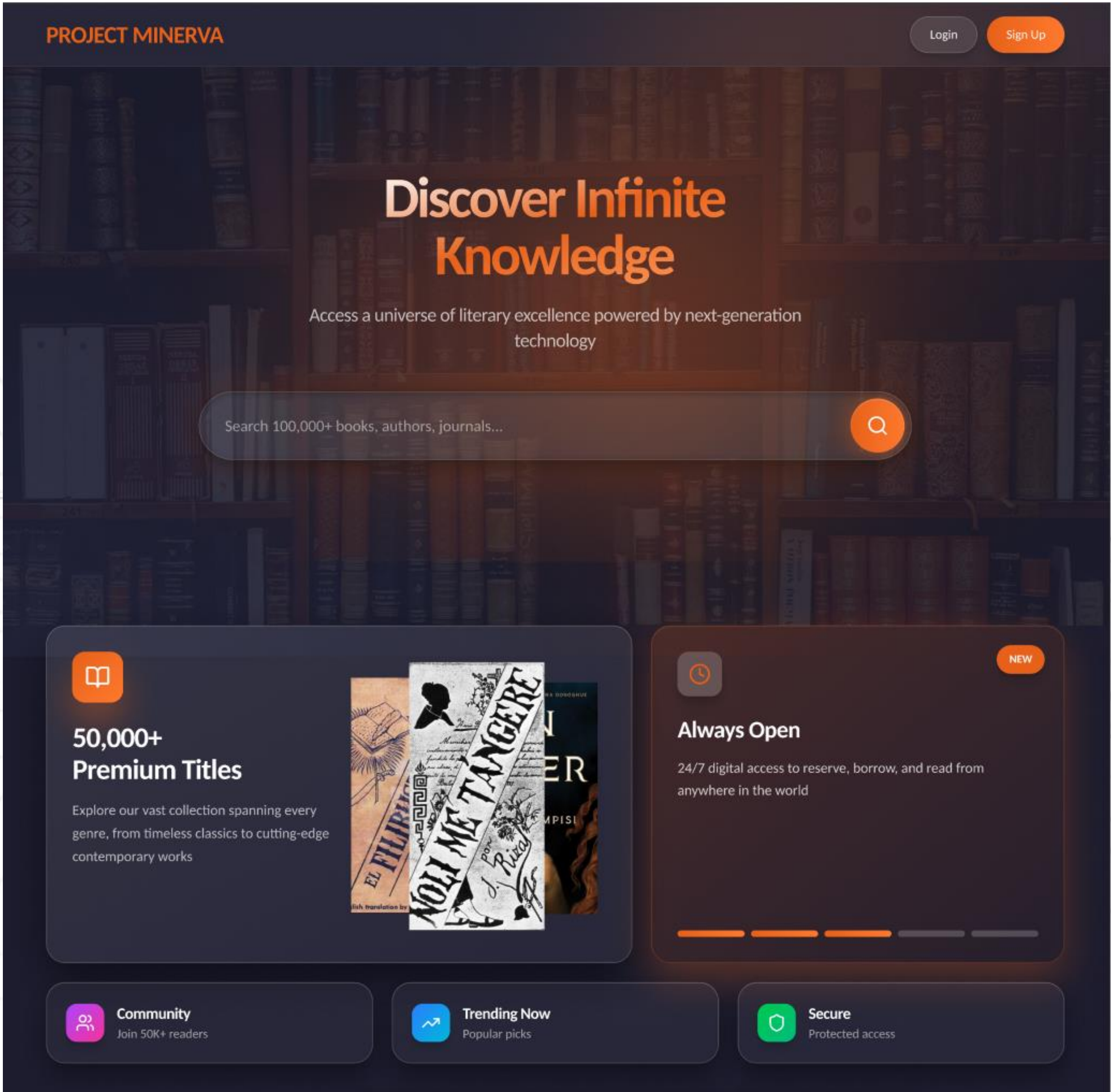
Modal List List

SCREEN	DESCRIPTION	RIGHTS / PERMISSIONS		REMARKS
		ADMIN	USER	
MDL-07 Clear Fine Confirmation	Final confirmation showing the total amount paid before restoring member privileges.	x		
MDL-08 Leave Review Modal	A modal allowing the user to leave a rating and review		x	

Screen Flow



SCR-01 Landing Page



SCR-01 Landing Page Logic

On "Search" Button Click

IF Search Keyword is entered

IF Matching Books Exist

- Redirect to Public Catalog View
- Display Matching Books

ELSE

- Stay on Landing Page
- Notification:
Text Label: **M009**

ELSE

- Display All Available Books

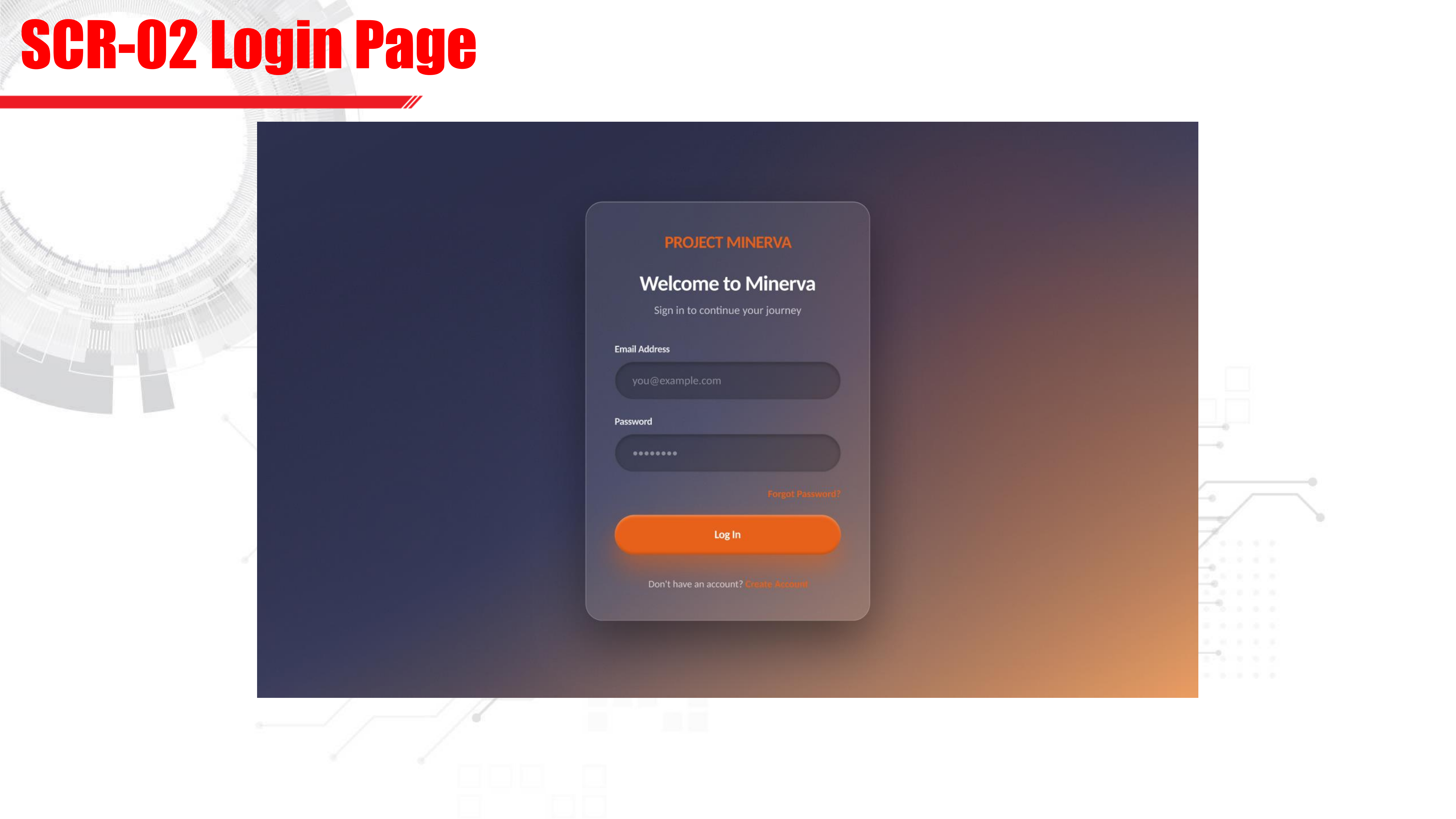
On "Log In" Button Click

- Redirect to Login Page (SCR-02)

On "Sign Up" Button Click

- Redirect to Signup Page (SCR-03)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.



SCR-02 Login Page

PROJECT MINERVA

Welcome to Minerva

Sign in to continue your journey

Email Address

you@example.com

Password

.....

[Forgot Password?](#)

Log In

Don't have an account? [Create Account](#)

SCR-02 Login Page Logic

On "Log In" Button Click

IF Email Address and Password fields are filled

IF Email Address format is valid

IF Account exists

IF Failed Login Attempts < 5

IF Credentials are correct

IF Role = Admin

- Redirect to Admin Dashboard Screen (SCR-10)

ELSE IF Role = Member

- Redirect to Member Catalog Dashboard (SCR-06)

ELSE

- Increment Failed Login Attempts

- Stay on Login Screen

- Notification:

Text Label: **M001**

ELSE

- Lock Account

- Send Account Lockout Notification Email

- Stay on Login Screen

- Notification:

Text Label: **M002**

ELSE

- Stay on Login Screen

- Notification:

Text Label: **M003**

ELSE

- Stay on Login Screen

- Notification:

Text Label: **M004**

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-02 Login Page Logic

On "Sign Up" Click - Redirect to Signup Screen (SCR-03)

ELSE

- Stay on Login Screen
- Notification:
Text Label: **M005**

IF Database is Down

- Redirect to Server Error Screen

On "Forgot Password" Click

- Redirect to Forgot Password Screen (SCR-04)

On "Sign Up" Click

- Redirect to Signup Screen (SCR-03)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-03 Signup Page

PROJECT MINERVA

Create Account

Join Minerva to start borrowing books

First Name

John

Last Name

Doe

Email Address

you@example.com

Password

.....

Confirm Password

.....

Create Account

Already have an account? [Sign In](#)

SCR-03 Signup Page Logic

On "Register" Button Click

IF Full Name, Email Address, and Password fields are filled

IF Email Address format is valid

IF Email Address does not exist

IF Password meets format requirements

- Create Member Account
- Send Signup Confirmation Email
- Redirect to Login Page
- Notification:
Text Label: **M006**

ELSE

- Stay on Signup Page
- Notification:
Text Label: **M007**

ELSE

- Stay on Signup Page
- Notification:
Text Label: **M008**

ELSE

- Stay on Signup Page
- Notification:
Text Label: **M004**

ELSE

- Stay on Signup Page
- Notification:
Text Label: **M010**

IF Database is Down

- Redirect to Server Error Screen

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-03 Signup Page Logic

On "Log In" Link Click

- Redirect to Login Page (SCR-02)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-04 Forgot Password Page

[← Back to Login](#)

PROJECT MINERVA

Reset Password

We'll send you a reset link via email

Email Address

 you@example.com

Send Reset Link

Remember your password? [Sign In](#)

SCR-04 Forgot Password Page Logic

On "Send Reset Link" Button Click

IF Email Address is filled

IF Email Address format is valid

IF Email Address exists

- Generate Password Reset Token
- Send Password Reset Email
- Redirect to Login Page
- Notification:
Text Label: **M011**

ELSE

- Stay on Forgot Password Page
- Notification:
Text Label: **M012**

ELSE

- Stay on Forgot Password Page
- Notification:
Text Label: **M004**

ELSE

- Stay on Forgot Password Page
- Notification:
Text Label: **M014**

IF Email Service is Unavailable

- Stay on Forgot Password Page
- Notification:
Text Label: **M013**

On "Back to Login" Click

- Redirect to Login Page (SCR-02)

On "Sign In" Click

- Redirect to SignUp Page (SCR-03)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-05 Change Password Page

[← Back to Login](#)

PROJECT MINERVA

Change Password

Enter and confirm your new password below

New Password



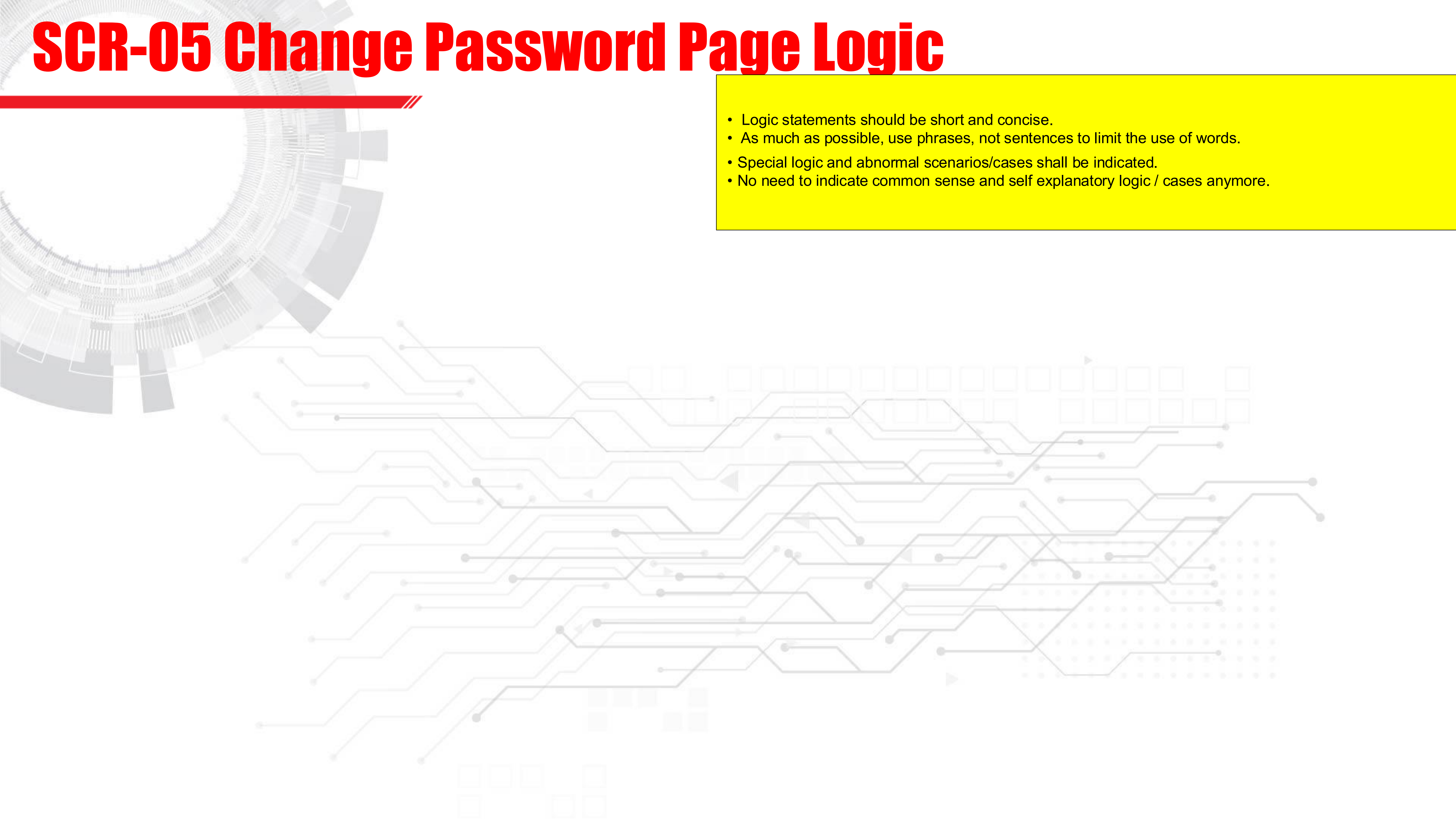
Confirm Password



Change Password

[Back to Login](#)

SCR-05 Change Password Page Logic



- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-06 Member Catalog Dashboard Logic

On "Search" Button Click

IF Search Keyword is entered

IF Matching Books Exist

- Display Matching Books

ELSE

- Stay on Member Catalog Dashboard
- Notification:

Text Label: **M009**

ELSE

- Display All Books

On Genre Selection

IF Matching Books Exist

- Display Filtered Books

ELSE

- Display Empty Result Set
- Notification:

Text Label: **M015**

On Book Card Click

- Redirect to Book Details Page (SCR-07)

On "My Loans" Click

- Redirect to Loans & History Page (SCR-08)

On "Profile" Click

- Redirect to User Profile & Settings Page (SCR-08)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-06 Member Catalog Dashboard

PROJECT MINERVA

CatalogMy LoansProfile

BROWSE

All

Fiction

Trending

Popular

New Arrivals

Classics

Good Morning, Sarah

Discover your next great read

Search by title, author, ISBN...



The Great Novel

Jane Smith



Science Explained

John Doe



History Chronicles

Mary Johnson



Mystery Tales

Bob Wilson



Poetry Collection

Alice Cooper



Adventure Stories

Mike Brown



The Secret Lives

Clara Osel



Leaf & Light

Tom Ridge

SCR-07 Book Details Page

PROJECT MINERVA

CatalogMy LoansProfile

← Back to Catalog



Available 14-day loan period

FICTION

The Great Novel

by Jane Smith

★★★★☆

4.5 / 5

A captivating story that takes readers on an unforgettable journey through time and space. This masterpiece explores the depths of human emotion and the complexities of modern society — weaving together multiple narratives across different eras, creating a rich tapestry of experience that resonates long after the final page.

#

ISBN

978-1234567890

📅

Published

2024

🏠

Publisher

Great Publishing House

Borrow This Book

☆ Leave a Review

★ Reader Reviews (127)

AM

Alex M.

★★★★★

Absolutely gripping from the first page. One of the best reads this year.

SR

Sofia R.

★★★★☆

Beautiful prose and complex characters. The ending left me speechless.

JT

James T.

★★★★★

A rare masterpiece. Highly recommend to anyone who loves deep storytelling.

SCR-07 Book Details Page Logic

On "Borrow" Button Click

IF Book Status = Available

IF Member Account is Active

IF No Outstanding Unpaid Fines

- Create Loan Record
- Update Book Status to Borrowed
- Update Available Quantity
- Send Borrow Confirmation Email
- Refresh Book Details Page
- Notification:

Text Label: **M016**

ELSE

- Stay on Book Details Page
- Notification:

Text Label: **M017**

ELSE

- Stay on Book Details Page
- Notification:

Text Label: **M018**

ELSE

- Stay on Book Details Page
- Notification:

Text Label: **M019**

IF Database is Down

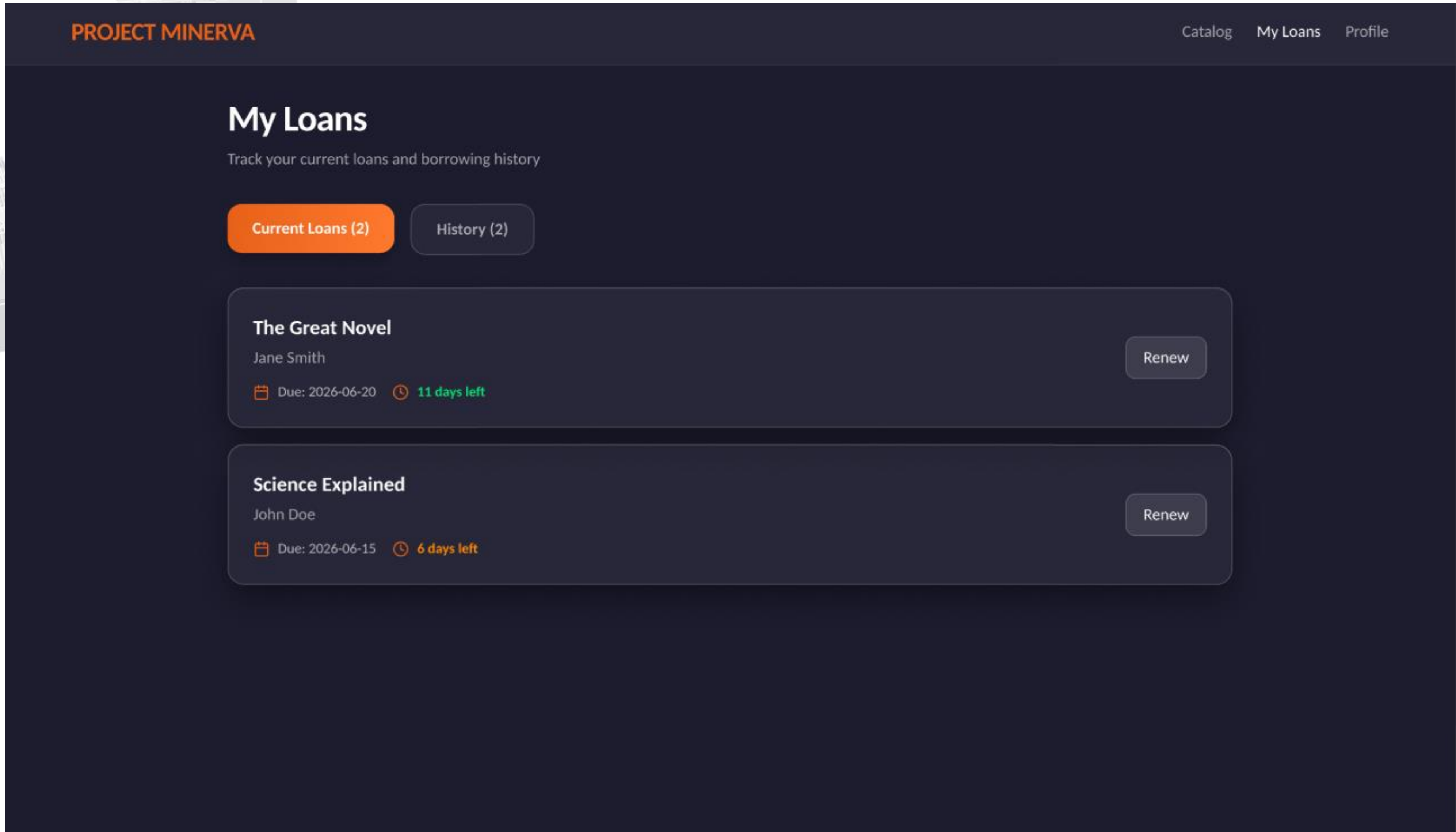
- Redirect to Server Error Screen

On "Back" Button Click

- Redirect to Member Catalog Dashboard (SCR-06)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-08 Loans & History Page



SCR-08 Loans & History Page Logic

On "Return Book" Button Click (Current Loans Tab)

Trigger MDL-02 (Return Confirmation Modal)

- Update Loan record ReturnDate
- Update Book Status to 'Available'

IF ReturnDate > DueDate

- Calculate Fine (Days Late * Daily Rate)
- Create Fine record (IsPaid = False)
- Suspend User Account
- Trigger Email: Account Suspension & Fine Invoice
- Notification: **M020**

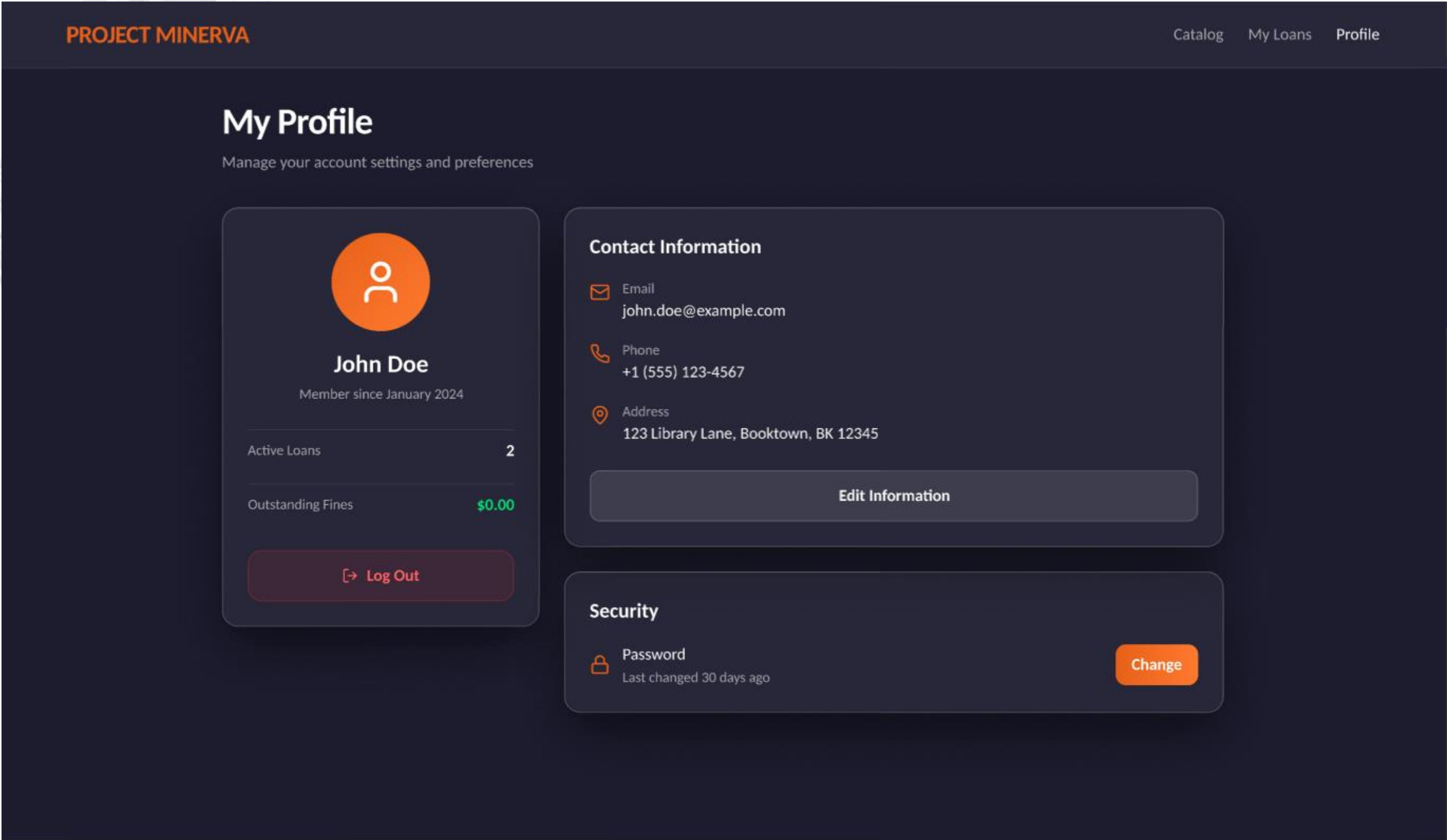
ELSE

- Trigger Email: Return Confirmation
- Notification: **M021**

Move Loan record to Borrowing History tab

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-08 User Profile & Settings



SCR-08 User Profile & Settings Logic

On "Change" Button Click

IF Current Password, New Password, and Confirm Password are filled

IF Current Password matches database

IF New Password matches Confirm Password

- Update PasswordHash in database
- Trigger Email: Password Change Confirmation
- Notification: **M022**

ELSE

- Notification: **M023**

ELSE

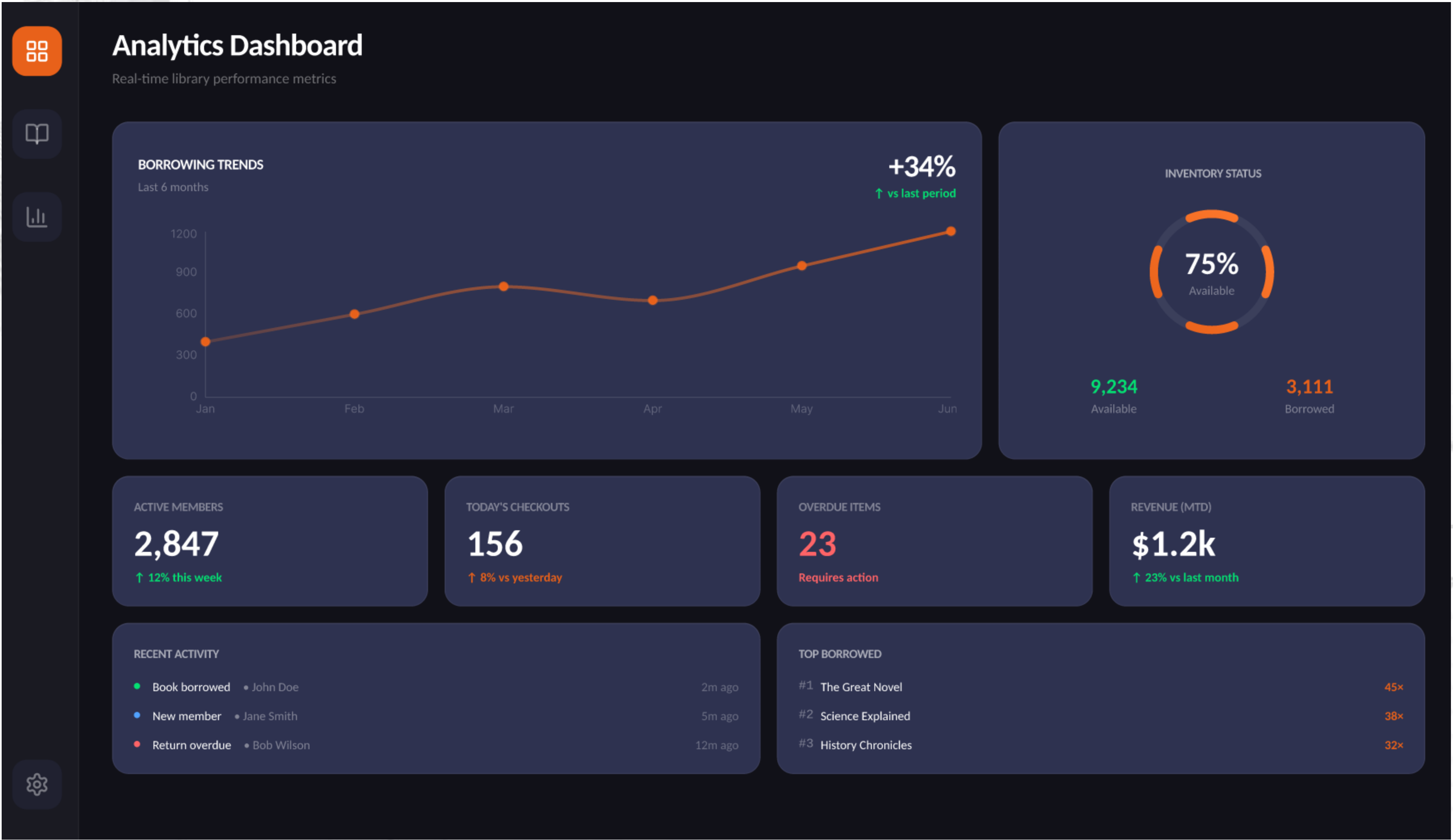
- Notification: **M024**

ELSE

- Notification: **M025**

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-10 Admin Dashboard



SCR-10 Admin Dashboard Logic

On Sidebar Navigation Click

- Redirect to selected module (SCR-11, SCR-13, SCR-14, or SCR-15)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-11 Book Management List

PROJECT MINERVA

Dashboard

Books

Members

Reports

Audit Trail

Book Inventory

12,345 total books

+ Add Book

Search by title, author, ISBN...

TITLE / AUTHOR	ISBN	GENRE	COPIES	STATUS	ACTIONS
The Great Novel Jane Smith	978 - 1234567890	Fiction	3/5	Available	
Science Explained John Doe	978 - 0987654321	Science	1/3	Available	
History Chronicles Mary Johnson	978 - 1122334455	History	4/4	Available	
Mystery Tales Bob Wilson	978 - 5544332211	Mystery	0/6	All Borrowed	
Lost Archives Sarah Connor	978 - 6677889900	Fiction	0/2	Lost	

Showing 1-5 of 12,345 books

Previous

1

2

Next

SCR-11 Book Management List Logic

On "Add New Book" Button Click

- Redirect to SCR-12 (Add / Edit Book Form) in 'Add' mode

On "Edit" Button Click (Row Level)

- Redirect to SCR-12 (Add / Edit Book Form) in 'Edit' mode passing BookID

On "Status Override" Button Click (Row Level)

- Trigger MDL-06 (Status Override Modal)

On Modal Confirm (MDL-06)

IF Book is currently 'Borrowed' AND new status is 'Archived'

- Prevent action
- Notification: **M037**

ELSE

- Update Book Status (Lost, Damaged, Archived)
- Create AuditLog record
- Refresh Grid

On "Bulk Import" Button Click

- Trigger MDL-05 (Bulk Import Upload Modal)

On File Upload Submit (MDL-05)

IF File is valid .csv format

IF CSV headers match the exact expected schema

- Parse CSV and insert into Books table (skip/log exact duplicate rows)
- Create AuditLog record
- Refresh Grid
- Notification: **M026** ELSE
- Notification: **M027** ELSE
- Notification: **M028**

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-12 Add / Edit Book Form

PROJECT MINERVA

Dashboard

Books

Members

Reports

Audit Trail

← Back to Book Management

Edit Book

Update book information and inventory

Book Cover

Upload File

Paste URL

Click to upload or drag and drop

PNG, JPG up to 5MB

Title *

Enter book title

Author *

Enter author name

ISBN *

978-1234567890

Publisher

Publisher name

Publication Year

2026

Total Copies *

5

Genre

Description

Enter book description...

Update Book

Cancel

SCR-12 Add / Edit Book Form Logic

On "Save Book" Button Click

IF required fields (Title, Author, Genre, Publication Year, Price) are filled

IF Publication Year $\leq$ Current Year AND Price $\geq$ 0

IF mode is 'Add'

- Insert new Book record
- Create AuditLog record
- Redirect to SCR-11 (Book Management List) ELSE IF mode is 'Edit'
- Update existing Book record
- Create AuditLog record
- Redirect to SCR-11 (Book Management List) ELSE
- Stay on Form Screen
- Notification: **M029**

ELSE

- Stay on Form Screen
- Notification: **M010**

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-13 System Audit Trail

PROJECT MINERVA

Dashboard

Books

Members

Reports

Audit Trail

Audit Trail

Admin actions only — read-only log

Export Logs

Search by action, admin, or details...

Timestamp	Action	Admin	Details
2026-06-13 09:12:04	Book Added	Admin Rivera	Added "The Midnight Garden" by Clara Osei
2026-06-12 16:44:31	Book Edited	Admin Rivera	Updated inventory count for "Science Explained" (3 → 5 copies)
2026-06-12 14:09:55	Loan Override	Admin Santos	Manually marked loan #1042 as returned — "History Chronicles"
2026-06-11 11:28:17	Book Deleted	Admin Rivera	Removed "Lost Archives" (all copies lost/unrecoverable)
2026-06-10 10:03:42	Account Restored	Admin Santos	Cleared fines & restored account for member Robert Chen
2026-06-09 08:55:11	Book Added	Admin Rivera	Added "Leaf & Light" by Tom Ridge

Showing 1-6 of 6 entries

Previous

1

Next

SCR-13 System Audit Trail Logic

On "Filter" Submit

IF Date Range or Admin ID is selected

IF Start Date <= End Date

- Filter AuditLog table results accordingly

IF results == 0

- Display Text Label: **M030**

ELSE

- Notification: **M031**

ELSE

- Display default full list (paginated)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-14 Reports Center

PROJECT MINERVA

[Dashboard](#)[Books](#)[Members](#)[Reports](#)[Audit Trail](#)

Reports Center

Generate and download comprehensive library reports

Generate New Report



Borrowing Activity Report

Track borrowing trends, popular books, and usage patterns over time

Start Date

End Date

Generate Report



Inventory Report

Complete overview of book stock, availability, and condition status

Start Date

End Date

Generate Report



Member Analytics

Member registration, engagement, and demographic statistics

Start Date

End Date

Generate Report



Overdue Books Report

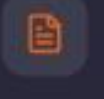
List of overdue items, fines, and follow-up actions required

Start Date

End Date

Generate Report

Recent Reports



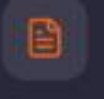
Monthly Borrowing Report - May 2026

2026-06-01

•

Borrowing

Download



Inventory Audit - Q2 2026

2026-05-28

•

Inventory

Download



Member Growth Report - May 2026

2026-05-15

•

Members

Download

SCR-14 Reports Center Logic

On "Generate Report" Button Click

IF Report Type is selected (Borrowing, Inventory, Member)

IF Date Range is valid (if applicable)

IF Start Date <= End Date

- Query database and render data table

IF Data Table has 1 or more records

- Enable "Export to CSV" button

ELSE

- Disable "Export to CSV" button

- Display Text Label: **M032**

ELSE

- Notification: **M031**

ELSE

- Notification: **M033**

ELSE

- Notification: **M034**

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

SCR-15 Member Management

PROJECT MINERVA

Dashboard

Books

Members

Reports

Audit Trail

Member Management

2 suspended accounts requiring attention

Search suspended members by name or email...

RC

Robert Chen

robert.chen@email.com

Suspended

PN

Priya Nair

priya.nair@email.com

Suspended

RC

Robert Chen

SUSPENDED

⚠ Overdue books — 42 days

Suspended on 2026-05-10

Email

robert.chen@email.com

Phone

+1 (555) 012-3456

Member Since

2023-03-15

Active Loans

2 books

👤

OUTSTANDING BALANCE

\$50.49

4 charges outstanding

BREAKDOWN

Overdue Fine — The Great Novel

\$12.50

Overdue Fine — Science Explained

\$8.00

Lost Book Fee — Mystery Tales

\$24.99

Processing Fee

\$5.00

Total

\$50.49

✔ Clear Fines & Restore Account

This action will waive all outstanding fines and reactivate the member's borrowing privileges.

SCR-15 Member Management Logic

On "Search User" Input Submit

IF Input is not empty

- Filter Member list by Name or Email

IF match is found

- Display outstanding fines in grid

ELSE

- Display Text Label: **M035**

On "Clear Fines" Button Click (User Profile view)

IF User has outstanding unpaid fines

- Trigger MDL-07 (Clear Fine Confirmation Modal) ELSE
- Button disabled/hidden

On Modal Confirm (MDL-07)

- Update all related Fine records to IsPaid = True
- Lift Account Suspension flag
- Trigger Email: Account Restored (Payment Receipt)
- Create AuditLog record
- Refresh Member Profile
- Notification: **M036**

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

Common – Logic / Behavior

Search 100,000+ books, authors, journals...



Catalog

My Loans

Profile

Dashboard

Books

Members

Reports

Audit Trail

Common – Logic / Behavior

Search Bar (SCR-01, SCR-06, SCR-11, SCR-13, SCR-15)

Search Keyword Input

IF Keyword entered

IF match found -> display matching results

IF no match found -> show M009

IF No Keyword entered -> display all records

Navigation Bar (SCR-06, SCR-07, SCR-08, SCR-08)

Catalog Link

On Click -> redirect to SCR-06 (Member Catalog Dashboard)

My Loans Link

On Click -> redirect to SCR-08 (Loans & History Page)

Profile Link

On Click -> redirect to SCR-08 (User Profile & Settings)

Admin Navigation Bar (SCR-10, SCR-11, SCR-12, SCR-13, SCR-14, SCR-15)

Dashboard Link

On Click -> redirect to SCR-10 (Admin Dashboard)

Books Link

On Click -> redirect to SCR-11 (Book Management List)

Audit Link

On Click -> redirect to SCR-13 (System Audit Trail)

Reports Link

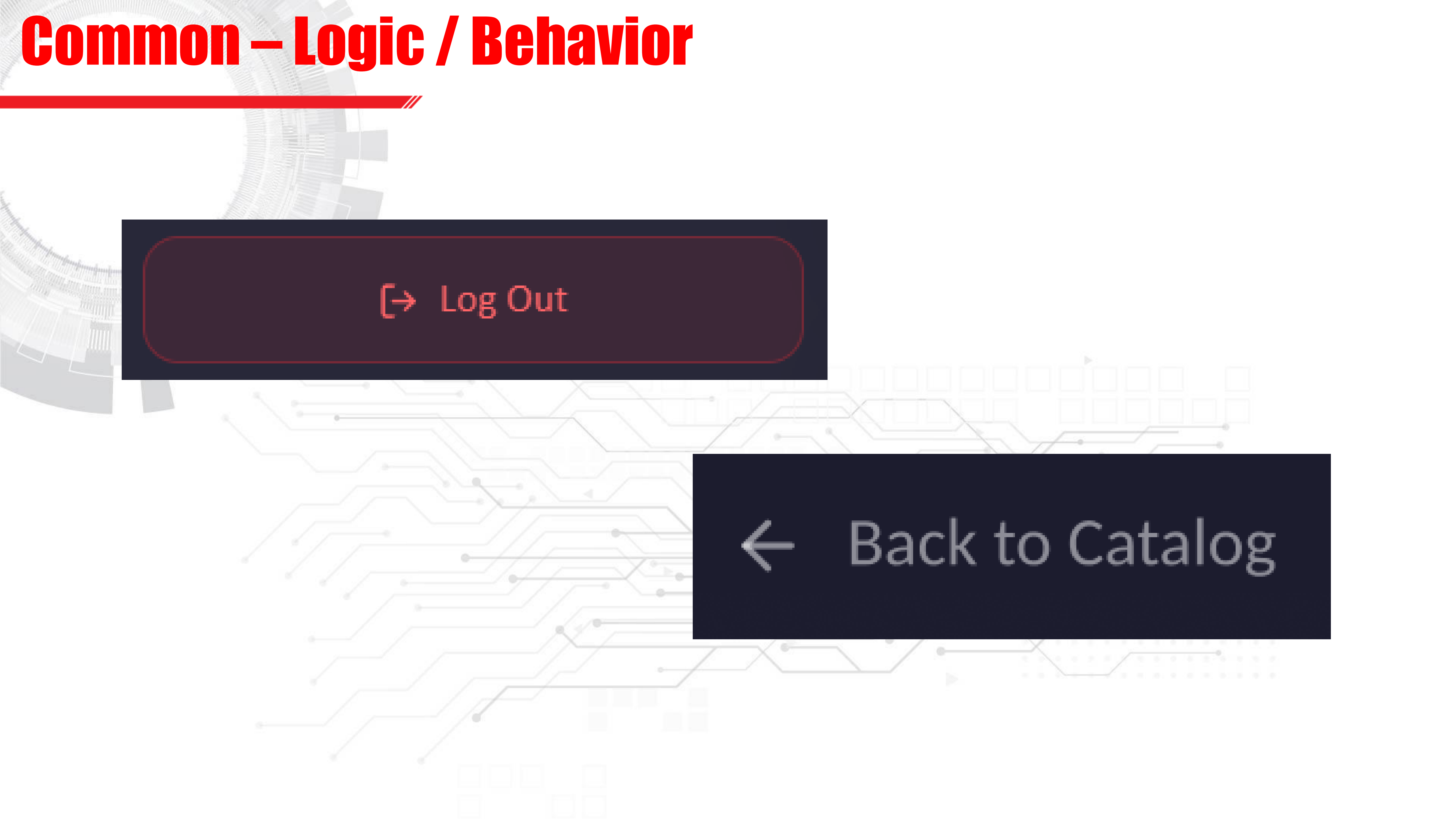
On Click -> redirect to SCR-14 (Reports Center)

Members Link

On Click -> redirect to SCR-15 (Member Management)

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

Common – Logic / Behavior



[→] Log Out

← Back to Catalog

Common – Logic / Behavior

Logout Button (All Authenticated Screens)

Logout Button

- On Click -> trigger MDL-03 (Logout Confirmation Modal)
- On confirm (Yes) → clear session → redirect to SCR-01 (Landing Page)
- On cancel (No) → close modal, stay on current screen

Back Button (SCR-07, SCR-12)

Back / Back to Catalog / Back to Book Management

- SCR-07 → On click → redirect to SCR-06 (Member Catalog Dashboard)
- SCR-12 → On click → redirect to SCR-11 (Book Management List)

Pagination (SCR-11, SCR-13, SCR-14)

Next / Previous / Page Number

- On click → load corresponding page of records
- New search or filter applied → reset to page 1
- Default page size: 5 records per page

Session Guard (All Protected Screens)

Unauthenticated Access

- IF no active session → redirect to SCR-02 (Login Page)
- IF Member accesses Admin screens (SCR-10 to SCR-15) → redirect to SCR-06
- IF Admin accesses Member screens (SCR-06 to SCR-08) → redirect to SCR-10
- IF Database is Down (any screen) → redirect to Server Error Screen

- Logic statements should be short and concise.
- As much as possible, use phrases, not sentences to limit the use of words.
- Special logic and abnormal scenarios/cases shall be indicated.
- No need to indicate common sense and self explanatory logic / cases anymore.

Common – Logic / Behavior

Dropdown fields

Property	Logic
Placeholder	If no default value is specified, show "select..." As placeholder. Otherwise, the default values should be pre-selected on load
Options order	If All is an option, it must appear first in the list. All other options are arranged in ascending alphabetical order .

• Date Fields Format

- Date – MM/dd/YYYY
- Datetime – MM/dd/YYYY hh:mm:ss where time is in (2H)
- Date Range – MM/dd/YYYY – MM/dd/YYYY

Common – Logic / Behavior

Input fields and Validation

Data type	Input validation
String	Accepts special characters. Max length defined per fields
Numeric - Integer	Does not accept special characters. Accepts whole numbers only. No decimals
Required String	If field is empty on submit, show inline error: M025
Timepicker	If the input value is invalid or empty, show inline error: M025
Email	Must watch valid email format user@domain.com . If invalid, show: M004
Password	Minimum 8 characters. If requirement not met, show: M007

Common – Logic / Behavior

Account states

State	Trigger	UI behavior	Resolution
Active	Default on registration	Full borrowing access	
Suspended	Unpaid fines exist (F-015)	Borrow button disabled. Inline notice: M017	Admin clears fine (F-035)
Locked	5 consecutive failed logins	Login blocked. Message: M002	Password reset (F-003)

- **Fine Calculation**

- **Overdue: Amount** = Days overdue * daily fine rate. **Days overdue** = ReturnDate – DueDate.
- **Lost book:** Amount = Books.Price (replacement cost). Applied when admin sets status to Lost (F-034)
- Fine row created in **Fines table** with **isPaid – false** on trigger event.
- A member may have multiple unpaid fines simultaneously
- Suspension activates if **any** fine for that user has **isPaid = false**
- Admin clears all outstanding fines in a single action (F-035) - not per fine individually

Common – Logic / Behavior

CSV Bulk Import Error Handling

Scenario	Behavior
File not .CSV	Reject immediately. Show :M028
File exceeds MB	Reject
Missing required columns	Reject entire file. List missing column names in error message
Individual row fails validation	Skip that row. Continue importing valid rows.
Duplicate book entry	Skip duplicate. Include in skipped count
Full import success	Show SUCCESS modal. Log to Audit Trail (F-036)

- **File Upload**
- Max file size: **MB**. Show error if file exceeds the maximum
- Accepted extensions vary per module. Filenames must support Unicode characters.
- Upload type – single or multiple depending on the module
- Failed uploads show an inline error; form must not submit partially.

Information Messages

MSG ID	MESSAGE	TYPE
M001	Invalid email or password	Error
M002	Account locked	Error
M003	Account does not exist	Error
M004	Invalid email format	Error
M005	Email address and password are required	Error
M006	Registration successful!	Success
M007	Password does not meet requirements	Error
M008	Email address already registered	Error
M009	No books found	Warning
M010	Required fields missing	Error
M011	Password reset sent link successfully	Success
M012	Email address not found	Error
M013	Unable to send reset email	Error
M014	Email address required	Error
M015	No books available in selected genre	Error
M016	Book borrowed successfully	Success
M017	Outstanding fines detected	Error

Information Messages

MSG ID	MESSAGE	TYPE
M018	Account suspended	Error
M019	Book not available	Warning
M020	Book returned late. Fines accrued	Warning
M021	Book successfully returned	Success
M022	Password successfully updated	Success
M023	New password do not match	Error
M024	Current password is incorrect	Error
M025	Please fill in all required fields	Error
M026	Import successful	Success
M027	Invalid CSV schema. Please check headers	Error
M028	Invalid file type. Please upload a .csv file	Error
M029	Invalid data: Check Year and Price values	Error
M030	No logs found for selected criteria	Warning
M031	Start date cannot be after end date	Error
M032	No records found for this period	Warning
M033	Please select a valid date range	Error
M034	Please select a report type	Error

Information Messages

MSG ID	MESSAGE	TYPE
M035	No member found matching criteria	Warning
M036	Fines cleared and account restored	Success
M037	Cannot archive a book that is currently checked out	Error



Thank You!

